

Experiment-16

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Course Code: -MLA0305

Slot: B

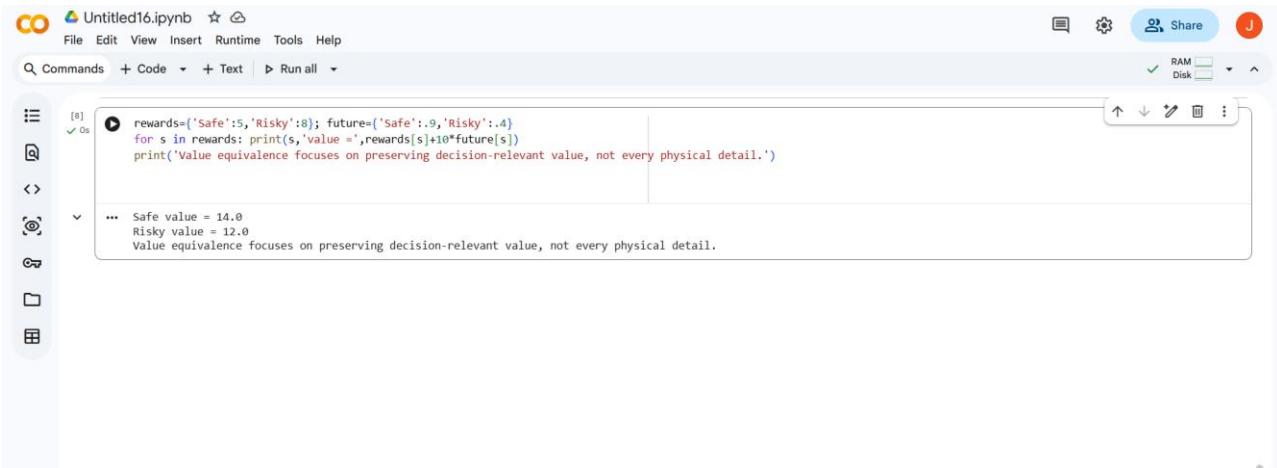
Title: Value-Equivalence Prediction for Model-Based Reinforcement Learning

Aim:

To analyze value-equivalence prediction techniques for improving planning accuracy and policy optimization in model-based Reinforcement Learning.

Procedure:

1. Define the environment and state-action space.
2. Collect transition samples from the environment.
3. Develop a predictive model for future states and rewards.
4. Estimate value functions using the learned model.
5. Compare predicted values with actual environment values.
6. Identify prediction errors.
7. Improve the model using additional training data.
8. Use the improved model for planning.
9. Evaluate policy performance.
10. Compare planning accuracy before and after model improvement.



The screenshot shows a Jupyter Notebook titled 'Untitled16.ipynb'. The code cell contains the following Python code:

```
rewards={'Safe':5,'Risky':8}; future={'Safe':.9,'Risky':.4}
for s in rewards: print(s,'value =',rewards[s]+10*future[s])
print('Value equivalence focuses on preserving decision-relevant value, not every physical detail.')
```

The output of the code cell is:

```
... Safe value = 14.0
Risky value = 12.0
Value equivalence focuses on preserving decision-relevant value, not every physical detail.
```

Result:

Value-equivalence prediction was used to improve planning accuracy. The improved model provided better value estimates and supported more effective policy optimization.